

Proof. Young's Inequality. Let $a, b \geq 0$ and let $p, q > 1$ satisfy $\frac{1}{p} + \frac{1}{q} = 1$. Then

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q},$$

with equality iff $a^p = b^q$ (including the trivial case $a = b = 0$).

1. Degenerate cases. If $a = 0$ or $b = 0$, the left-hand side is 0 while the right-hand side is a sum of non-negative terms; thus the inequality holds, and equality occurs only when $a = b = 0$. Hence we may assume $a > 0$ and $b > 0$.

2. Weighted AM–GM. Set

$$\alpha = \frac{1}{p}, \quad \beta = \frac{1}{q},$$

so that $\alpha, \beta > 0$ and $\alpha + \beta = 1$ because $\frac{1}{p} + \frac{1}{q} = 1$.

The weighted arithmetic–geometric mean inequality states that for any $X, Y \geq 0$,

$$\alpha X + \beta Y \geq X^\alpha Y^\beta, \quad (2.1)$$

with equality iff $X = Y$.

Choose $X = a^p$ and $Y = b^q$. Then

$$X^\alpha = (a^p)^{1/p} = a, \quad Y^\beta = (b^q)^{1/q} = b,$$

and (2.1) becomes

$$\frac{a^p}{p} + \frac{b^q}{q} \geq ab. \quad (2.2)$$

Inequality (2.2) is precisely Young's inequality.

3. Equality condition. Equality in (2.1) holds exactly when $X = Y$, i.e. when $a^p = b^q$. Under this relation the substitutions above give $\frac{a^p}{p} + \frac{b^q}{q} = ab$, so equality is attained in Young's inequality. Conversely, if equality holds in Young's inequality, then (2.2) must be an equality in the weighted AM–GM step, hence $a^p = b^q$. The trivial case $a = b = 0$ also satisfies $a^p = b^q$.

4. Conclusion. For all non-negative a, b and conjugate exponents $p, q > 1$,

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q},$$

with equality iff $a^p = b^q$ (including $a = b = 0$). □ □